PROXIMA

Built for classrooms
Not just meetings

Problems:

The Problem — "The Engagement Gap"

The primary hurdle in digital education is the "Communication Barrier." Current platforms force a choice between silence and disruption.

- Passive Learning: Students hesitate to provide feedback because unmuting or typing in chat interrupts the lecture's natural flow.
- Instructional Blindness: Educators lack real-time visibility into student comprehension, often realizing a concept wasn't understood only after the session ends.
- Dashboard Fragmentation: Managing multiple sessions, breakout rooms, and external tools leads to "Tab Fatigue," causing teachers to lose focus on their students.

Solutions:

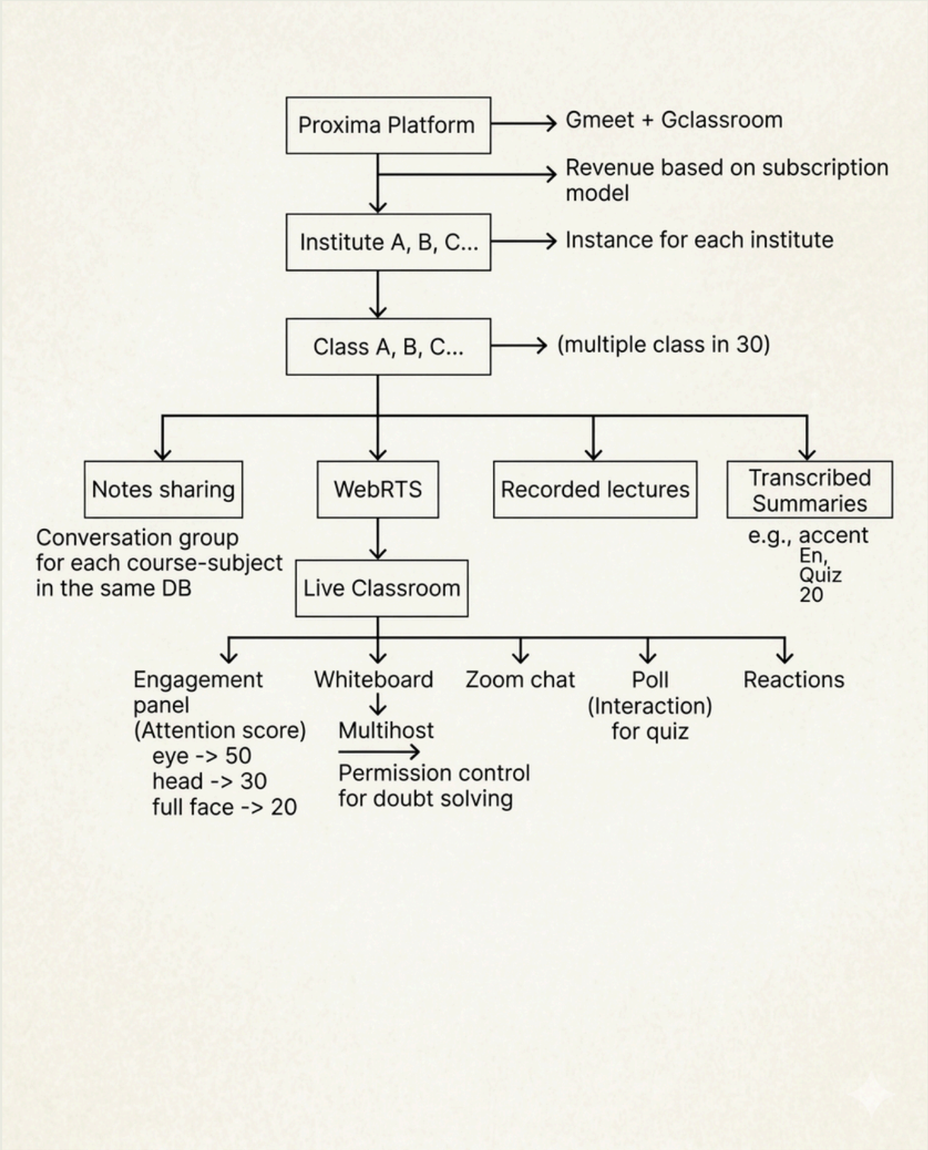
Our Solution — "The Proxima Architecture"

Proxima is a high-performance, Unified Engagement Layer designed to turn one-way broadcasts into two-way interactive experiences.

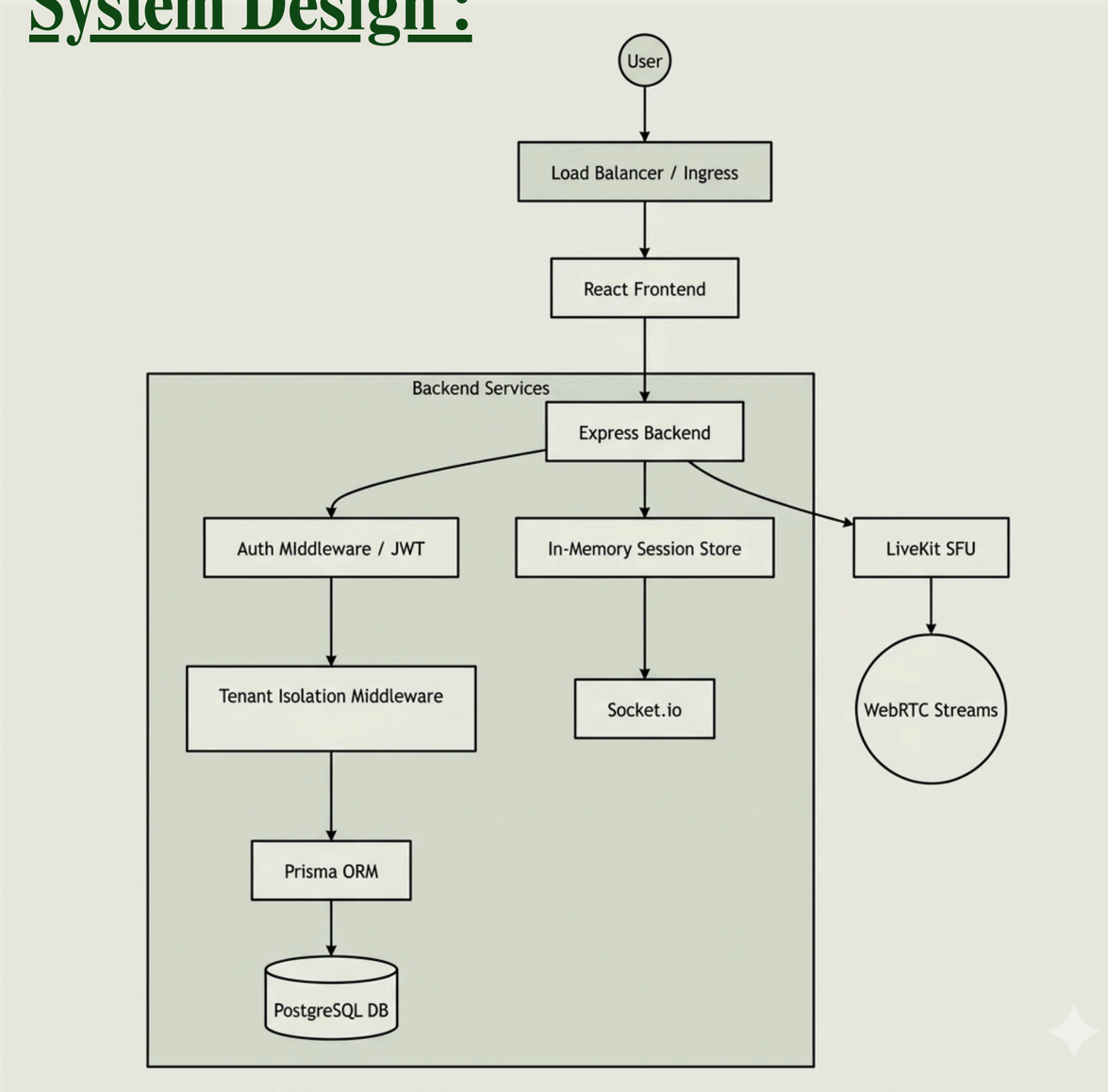
- Low-Latency Infrastructure: By leveraging WebRTC Data Channels, Proxima establishes a direct peer-to-peer pipeline for instant, non-verbal feedback that bypasses heavy video/audio processing.
- Centralized Orchestration: We provide a Unified Multi-Session Interface that consolidates every classroom tool—video, reactions, and assessments—into a single-pane-of-glass dashboard.
- Active Feedback Loops: Proxima bridges the "Validation Gap" by integrating assessment tools directly into the live stream, allowing for real-time course correction.

Team Name : - TeamTaam
Team Members : - Animesh, Akshita, Abhishek, Arpan

Product Design :



System Design :

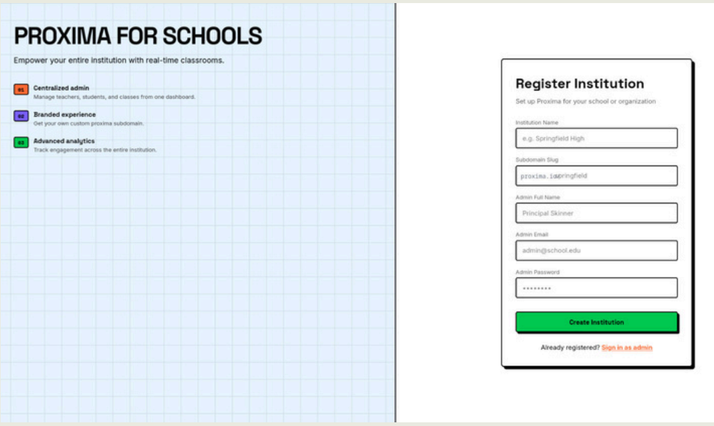


Technical & Implementation Feasibility

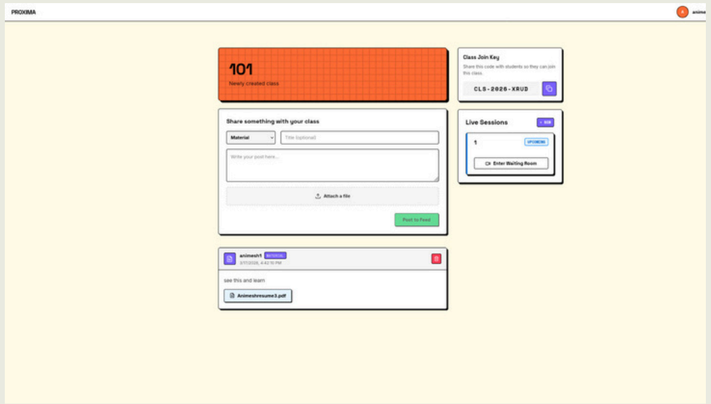
- Built for Speed: Leverages WebRTC and SFU-based architecture (LiveKit) for sub-500ms latency, ensuring a "live" feel for both video and data reactions.
- Scalable by Design: A PostgreSQL + Row-Level Security (RLS) backend ensures secure tenant isolation, allowing seamless support for multiple institutions.
- Frictionless Adoption: A lightweight Join Key System removes onboarding hurdles, while a Modular Architecture allows for a phased rollout (Classroom → Feed → AI Analytics).
- Cost-Optimized AI: Incremental integration of Whisper and LLMs ensures that high-compute features like auto-summarization don't bloat initial overhead.

Tech Stack

Working Prototype



Registration window



Class creation

Inside room

Challenges

- **Challenge: Real-Time Scalability & Latency**
  - The Risk: Concurrent live sessions with hundreds of students can spike server CPU and increase "lag," ruining the interactive experience.
  - The Proxima Solution: Implementing an SFU-based architecture (LiveKit). Unlike traditional mesh networks, an SFU handles media routing efficiently. We utilize Horizontal Auto-Scaling and Load Balancing to spin up new instances dynamically as student traffic increases.
- **Challenge: AI Processing Latency**
  - The Risk: Large audio files from 60-minute lectures can take several minutes to transcribe, delaying the "Instant Summary" feature.
  - The Proxima Solution: We employ Chunk-Based Stream Processing. Audio is sliced into 30-second buffers for "rolling" transcription. Summaries are generated in intervals (e.g., every 10 minutes) and finalized post-session to ensure 99% accuracy without lagging the live feed.

Value & Impact

- Proxima improves online education by enabling real-time interaction, engagement tracking, and seamless class management using WebRTC.
- It increases efficiency for teachers, reduces cost by replacing multiple tools with a single SaaS platform, and improves accessibility through simple join-key based entry.
- The platform introduces innovation with AI-powered attention detection and a subscription-based SaaS model for institutions, ensuring scalability and sustainable growth.

STAKEHOLDER	CORE VALUE PROPOSITION
Students	Eliminates the "fear of interrupting" with low-latency WebRTC reactions and provides instant AI-generated study guides.
Teachers	Provides a "Single-Pane-of-Glass" dashboard to manage sessions and track student engagement levels in real-time.
Institutions	Reduces overhead by replacing 3–4 fragmented tools (Zoom, PollEverywhere, Otter.ai) with one secure, Multi-Tenant SaaS.
Proxima Platform	Built on a scalable Subscription Model with high-margin AI features, ensuring long-term sustainable growth.

Our Future Path

High-Fidelity Academic Learning

- Live K-12 Classrooms: Enables teachers to track real-time engagement and "confusion levels" through WebRTC reactions without stopping the lesson.
- Higher Education & Universities: Ideal for large college lectures where Attendance Monitoring and Attention Heatmaps provide instructors with deep data on student retention.
- Hybrid Classroom Integration: Bridges the gap between offline and online students, allowing remote participants to feel as present and interactive as those in the physical room.

Targeted Instruction & Training

- Coaching & Exam Prep Institutes: Conduct high-intensity interactive batches where instant quizzes and polls help students prepare for competitive exams with immediate feedback.
- Personalized Remote Tutoring: Enhances 1-on-1 sessions with Live Annotation and a dedicated Doubt-Solving Channel to ensure no student is left behind.
- Corporate Upskilling: Tailored for fast-paced training sessions with instant feedback loops and certification-ready quiz results.

Large-Scale Management & Insights

- Multi-Class Orchestration: Empowers teachers handling different subjects or sections to manage multiple live sessions and breakout rooms from a single, unified dashboard.
- Massive Open Webinars: Scales to large audiences while maintaining high-fidelity interactivity; organizers get granular Engagement Insights instead of just a participant count.
- AI-Driven Post-Session Analytics: Automatically generates summaries and "Action Items" for every session, making the content searchable and reusable for all stakeholders.

THANK-YOU